FetalAI: head circumference report

HC18 U-Net segmentation · research prototype 11/13/2026, 11:53:28 AM

Academic research prototype, not a medical device. Not clinically validated; must not be used for diagnosis or pregnancy-care decisions.

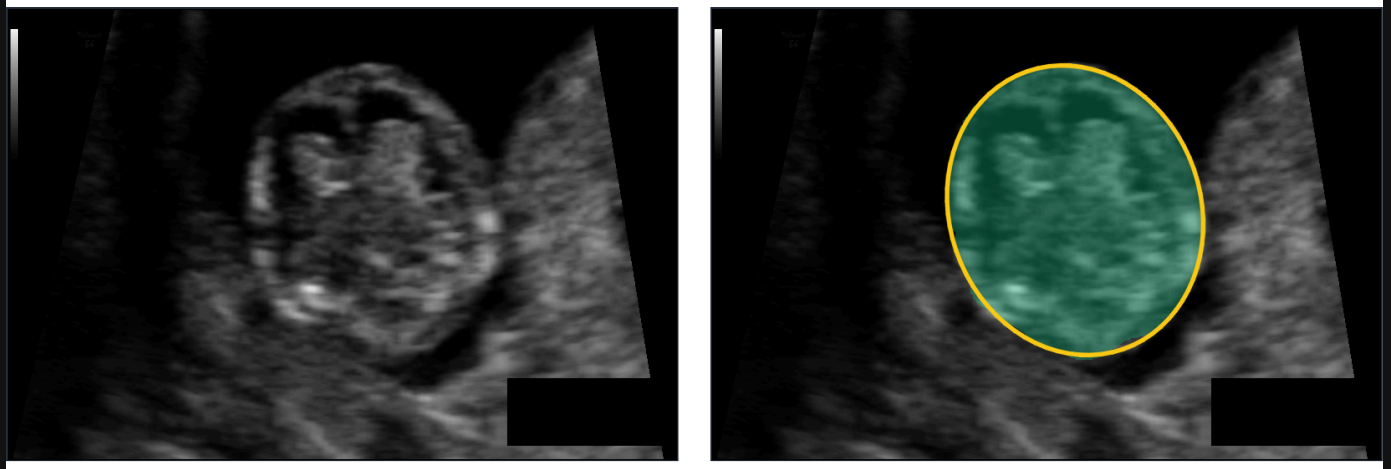


Image	006_HC.png (800 × 540 px)
Pixel spacing	0.06560565 mm/px
Plane check	WARN (overridden): This looks like a trans-cerebellum brain plane, not trans-thalamic. HC is defined on the trans-thalamic plane; please re-acquire. Measured anyway because override_plane_check was set.
Head detected	yes
Head circumference	67.3 mm ± 0.3 mm (SD over 8 TTA passes)
Reliability	OK (CV 0.52%, disagreement 6.5%)
Ellipse diameters	23.1 × 19.7 mm
Head area	3.59 cm²
Model confidence	97.9%
Model version	hc18-unet-epoch50

Screening summary: Review recommended

Plane check: Trans-thalamic plane not confirmed (measured with override); re-acquire (R4)

Not assessed: CTG (no CTG features submitted); HC centile (needs HC in mm and gestational age).

The ultrasound and CTG models were trained on different patients (HC18 / FETAL_PLANES_DB vs UCI CTG); there is no paired data, so these fusion rules are illustrative and the combined level is not calibrated. Learned fusion and its calibration are Phase-2 work requiring paired clinical data.

Method: U-Net segmentation, ellipse fit, perimeter by Ramanujan's approximation × pixel spacing. Trained on the HC18 dataset (van den Heuvel et al., PLoS ONE 2018).